

Working with Multiple Open-Weight LLMs Using Ollama

- Why Use Multiple LLMs?
 - ✓ Different LLMs are designed for different use cases.
 - ✓ No single LLM is the best for every application.
 - ✓ AI applications can switch between models based on their requirements.

- Since we are currently working with Open-Weight LLMs using Ollama, you can explore all the supported Open-Weight LLMs from the official Ollama Model Library.
- Official Ollama Model Library:
<https://ollama.com/library>
- How to Evaluate and Select an LLM ?
 - Use the same prompt with different LLMs.

Open-Weight LLMs in Ollama

Different Models, Different Strengths, Different Purposes

General Purpose	Coding	Reasoning
General Purpose Chatbots, Q&A, content generation	Coding Code generation, debugging, code explanation	Reasoning Logical reasoning, math, complex problem solving
Best for: - General conversations - Q&A systems - Content creation - Everyday AI assistants	Best for: - Code generation - Debugging & fixing issues - Code explanation - Developer productivity	Best for: - Logical reasoning - Mathematics - Complex problem solving - Multi-step thinking tasks
Lightweight	Multilingual	Enterprise
Lightweight Fast inference on laptops and edge devices	Multilingual Applications supporting multiple languages	Enterprise Business applications, document processing, internal AI assistants
Best for: - Low-resource devices - Fast and efficient inference - Edge deployments - Education & experimentation	Best for: - Multilingual chatbots - Translation - Global applications - Cross-language understanding	Best for: - Business automation - Document processing - Internal AI assistants - Secure enterprise deployments

Key Takeaway: There is no single "best" model for everything. Choose the right model based on your use case, hardware, speed, quality, and resources.

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Hardware Guidelines for Open-Weight LLMs (Ollama)



These are rules of thumb, not strict requirements. Actual needs vary by model, quantization (e.g., Q4, Q5, Q8) and your use case.

General Hardware Guidelines by Model Size

Model Size (Parameters)	Recommended RAM (Minimum)	Typical Usage	Storage (Model Size) (Approx.)	Notes
 1B – 3B	8 GB 	Lightweight, laptops 	~1 GB – 3 GB 	✓ Best for basic tasks, fast and efficient
 7B – 8B	16 GB 	General-purpose development 	~4 GB – 8 GB 	✓ Good balance of quality and speed
 13B – 14B	24 – 32 GB 	Better reasoning and quality 	~8 GB – 16 GB 	✓ Improved reasoning and accuracy
 30B – 34B	48 – 64 GB 	High-end workstations or servers 	~20 GB – 40 GB 	✓ For advanced tasks and complex workloads
 70B+	128 GB+ RAM or GPUs 	Production servers with GPUs 	~40 GB+ 	✓ Best performance with high-end GPUs



CPU Requirements

There isn't a separate CPU requirement for each model.

- ✓ Modern 64-bit processor
- 4 At least 4 CPU cores
- 8+ 8 or more cores recommended for larger models
- ★ More CPU cores generally improve inference speed, but the required RAM is often the primary limiting factor.



GPU Requirements

If you have:



NVIDIA GPU



Apple Silicon GPU



Ollama can use GPU acceleration (where supported), making inference significantly faster.



Larger models benefit much more from GPUs than smaller ones.



Quick Tips



Choose a model size that matches your hardware.



Ensure you have enough RAM first – it is the most critical requirement.



Use GPU (NVIDIA / Apple Silicon) for faster and smoother performance.



Higher quantization (e.g., Q4) reduces memory usage but may lower quality.



Remember: Start with a smaller model, test on your system, and scale up based on your hardware and performance needs.

Working with Multiple Open-Weight LLMs Using Ollama...

 **Ollama**

Installing Multiple Open-Weight LLMs Using Ollama

Install different LLMs for different use cases and switch between them anytime!

1 General Purpose LLM

 **Model:**
Llama 3.2

Purpose:

- General conversations
- Q&A
- Content generation
- AI assistants

Install Command:

```
ollama pull llama3.2
```

2 Lightweight LLM

 **Model:**
Gemma 3

Purpose:

- Lightweight AI applications
- Fast inference
- Laptops and low-resource systems

Install Command:

```
ollama pull gemma3
```

3 Coding LLM

 **Model:**
DeepSeek-R1

Purpose:

- Code generation
- Debugging
- Code explanation
- Programming assistance

Install Command:

```
ollama pull deepseek-r1
```

4 Reasoning LLM

 **Model:**
Qwen 3

Purpose:

- Logical reasoning
- Mathematics
- Problem solving
- Analytical tasks

Install Command:

```
ollama pull qwen3
```

5 Multilingual LLM

 **Model:**
Qwen 3

Purpose:

- Multiple languages
- Translation
- Multilingual chatbots

Install Command:

```
ollama pull qwen3
```

Note: Qwen 3 is a versatile model and is commonly used for both reasoning and multilingual tasks.

6 Enterprise LLM

 **Model:**
Granite 3.3

Purpose:

- Enterprise AI
- Document processing
- Internal AI assistants
- Business applications

Install Command:

```
ollama pull granite3.3
```

Verify Downloaded Models

```
$ ollama list
```

Example Output:

NAME	ID	SIZE	MODIFIED
llama3.2	a00c4f17acd5	2.9 GB	2 hours ago
gemma3	8a6502fb9b0c	3.3 GB	3 hours ago
deepseek-r1	8a6502fb9b0c	4.7 GB	4 hours ago
qwen3	4a221a3c1f92	5.2 GB	5 hours ago
granite3.3	4961694c72d8	6.1 GB	6 hours ago

View Detailed Information About a Model

```
$ ollama show llama3.2
```

Example Output (Partial):

```
Model architecture: llama
parameters: 3.2B
context length: 8192
license: llama3.2
size: 2.9 GB
digest: a00c4f17acd5...
```

Manage and Control Models

View Currently Running Models

```
$ ollama ps
```

Example Output:

NAME	ID	SIZE	PROCESSOR	USAGE
llama3.2	a00c4f17acd5	2.9 GB	CPU	10s
qwen3	4a221a3c1f92	5.2 GB	CPU	10s

Shows the models that are currently loaded and running.

Stop a Running Model

```
$ ollama stop llama3.2
```

Stops the running model (llama3.2 in this example).

Remove a Downloaded Model

```
$ ollama rm llama3.2
```

Removes the model from your local system.

Check Ollama Version

```
$ ollama --version
```

Example Output:

```
ollama version 0.5.9
```

View All Commands

```
$ ollama help
```

Displays all available Ollama CLI commands.

How It Works

1. Install Multiple Models
Download any number of Open-Weight LLMs.
2. All Models Stored Locally
Ollama manages and stores them efficiently.
3. Switch Anytime
Change the model name in your request or command.
4. Use in Your Application
No changes to the rest of your code are required.

Key Takeaway

Different models for different needs. Install once, switch anytime, use easily!